

Data Displays and Shapes — Mixed Review

Answer Key

Grade 3 — Data, Measurement and Geometry

Quick Answers

- | | |
|--------------|---|
| 1. 6 | 8. 14 shapes |
| 2. 22 shapes | 9. (a) how many of the four shapes = 2, — |
| 3. 5 | 10. 1 |
| 4. 4 | 11. 4 |
| 5. 17 shapes | 12. (a) four-sided cut-outs after the change = 7, — |
| 6. 1 | |
| 7. 5 | |
-

What is verified

10 of 12 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 9, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 3 — Data, Measurement and Geometry

3.G.A – *problems 4, 5, 6, 8, 9*

3.MD – *problems 1, 2, 3, 7, 10, 11, 12*

Difficulty 1–5 of 5 across 14 tagged checks.

Common wrong answers

- 5. *If they got 22: counted every group in the graph, including the three-sided shapes*
 - 6. *If they got 2: assumed every four-sided shape has two pairs of parallel sides*
 - 8. *If they got 17: included the trapezoids, whose corners are not square corners*
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Bar Graph A: squares 8, rectangles 6, triangles 5, trapezoids 3.

Line Plot B: 3 sides has 4 marks, 4 sides has 5, 5 sides has 1.

Shape Set: P is a square, Q a rhombus, R a rectangle, T a trapezoid.

1. Bar Graph A: how many rectangles?

Solution: Find the rectangle bar and read its height against the scale on the left.

6

2. Bar Graph A: how many shapes altogether?

Solution: Every bar counts shapes, so add all four bar heights: $8 + 6 + 5 + 3 = 22$.

22 shapes

3. Bar Graph A: how many more squares than trapezoids?

Solution: “How many more” compares two bars, so subtract the shorter from the taller: $8 - 3 = 5$.

5

4. Shape Set: square corners on shape R?

Solution: Shape R is a rectangle. Test each corner with the corner of a card — all four fit.

4

5. Bar Graph A: how many quadrilaterals?

Solution: Two steps, and the first is a shape decision, not a counting one: squares, rectangles and trapezoids all have four straight sides, so they are quadrilaterals; triangles have three and are not. Then add only those three bars: $8 + 6 + 3 = 17$. Adding all four bars gives 22 and answers a different question.

17 shapes

6. Shape Set: pairs of parallel sides on shape T?

Solution: Shape T is a trapezoid. Its top and bottom sides stay the same distance apart — one pair. The two slanted sides lean towards each other, so they are not parallel.

1

7. Line Plot B: how many cut-outs have 4 sides?

Solution: Find 4 on the scale of Line Plot B and count the marks stacked above it.

5

8. Bar Graph A: how many shapes have four square corners?

Solution: Decide the group first: squares and rectangles have four square corners; trapezoids do not (their corners are two sharp and two blunt), and triangles do not have four corners at all. Then add just those two bars: $8 + 6 = 14$.

14 shapes

9. Shape Set: (a) how many shapes have four equal sides? (b) Why is having four sides not enough?

Solution (a): Measure the sides of each shape. P (square) has four equal sides and Q (rhombus) does too. R (rectangle) has two long sides and two short ones; T (trapezoid) has four different-looking sides. That is two shapes.

2

Solution (b) — instructor-judged. A full-credit answer says the group is about the sides being the same *length*, not about how many there are, and gives a shape that has four sides but fails: the rectangle or the trapezoid. Model answer: “All four shapes have four sides, but the group needs all four sides the same length. The rectangle’s sides are two long and two short, so it is out.” Repeating the number two is not full credit.

10. Bar Graph A: how many more rectangles than triangles?

Solution: $6 - 5 = 1$. The two bars look close, which is exactly why they should be read against the scale rather than compared by eye. 1

11. Line Plot B: how many more quadrilateral cut-outs than five-sided cut-outs?

Solution: The quadrilateral cut-outs are the ones above 4 on the scale — five of them. The five-sided cut-outs are above 5 — one of them. $5 - 1 = 4$. 4

12. Line Plot B: two more square cut-outs are made. (a) How many cut-outs have 4 sides then? (b) What happens to the tallest column?

Solution (a): A square has four sides, so both new cut-outs add marks above 4: $5 + 2 = 7$. 7

Solution (b) — instructor-judged. A full-credit answer says the column above 4 grows from five marks to seven and is now clearly the tallest, and explains that the column above 3 is untouched because the new cut-outs are four-sided. Model answer: “The 4 column goes from 5 marks to 7, so it is still the tallest and now further ahead. The 3 column stays at 4 marks, because a square has four sides and adds nothing to the three-sided group.” Giving 7 without saying why the three-sided column is unchanged is not full credit.